Reservoirs and wetlands

Overview

Reservoirs are standalone water body objects in the routing network. They receive inflow from upstream channels, HRUs, or recall objects, store it, and release it according to a decision table in `res_rel.dtl`. Each reservoir runs sediment settling, in-pond nutrient processes, and optional pesticide, salt, and constituent processes.

Wetlands are a separate object type linked to individual HRUs. They sit on the HRU surface, intercept surface runoff before it reaches the channel, and run a similar water balance and sediment-nutrient set as reservoirs. Wetland storage is configured per HRU in `wetland.wet`.

Process equations

Reservoir water balance

The daily water balance for a reservoir in `res_control.f90` is

```
stor_new = stor_prev + inflow + precip - outflow - evap - seep - irrig
```

where:

- `inflow` is the upstream hydrograph (`ht1`).
- `precip` is precipitation onto the water surface from the assigned weather station, using the current surface area.
- `outflow` is the release computed by `res_hydro.f90`.
- `evap` is potential evaporation times the water surface area, optionally adjusted by the reservoir evaporation coefficient.
- `seep` is seepage through the bed, computed from the bed hydraulic conductivity over the surface area.
- `irrig` is water withdrawn for irrigation through the water allocation module.

Surface area as a function of storage is interpolated from the principal volume `pvol_m3` and emergency volume `evol_m3` defined for the reservoir.

Release rules via `res_rel.dtl`

Releases are not hard-coded. Each reservoir points to a decision table in `res_rel.dtl`. The table lists conditions (volume above principal, day of year, downstream demand, …) and corresponding actions (weir release, target release, percent of volume, irrigation supply, …). At each daily step `res_hydro.f90`:

1. Walks the action list (`d_tbl%acts`).
2. For each action, evaluates whether all condition alternatives are satisfied (`d_tbl%act_hit`).
3. Fires the corresponding release equation.

Release types include:

- `weir_release` for a weir of height `wbody_prm%weir_hgt` and width `wbody_prm%weir_wid`. The discharge over time follows the weir equation with the daily volume above the weir crest as the head source.
- `pvol_release` and `evol_release` for releases triggered when storage exceeds the principal or emergency volume.
- Doell-style storage-based release with coefficient `kr` and exponent `alpha` (Jose T 2025 extension in `res_hydro.f90`).
- Hanazaki monthly-target release using monthly inflow and demand.
- HYPE seasonal sine and linear storage modulation for hydropower-style operations.

The weir branch calls `res_weir_release.f90` to integrate over sub-daily steps.

Reservoir sediment

`res_sediment.f90` computes settling using a first-order rate. If the in-reservoir sediment concentration (ppm) exceeds the equilibrium concentration `wbody_prm%sed%nsed`, the excess is reduced by the settling coefficient `wbody_prm%sed_stlr_co`:

```
sed_ppm_new = (sed_ppm - nsed) * sed_stlr_co + nsed
```

After settling, sand, large aggregates, silt, clay, and gravel are recomputed; all of the sand-sized fraction is assumed to deposit. Outflow sediment leaves at the post-settling concentration.

Reservoir nutrients

`res_nutrient.f90` runs an in-pond nutrient set: settling of organic N and organic P, mineralisation, nitrification, and algal growth limited by light and nutrients. Soluble nutrients track storage-weighted concentration and leave at the outflow concentration. Settling rates are read from `nutrients-res.nut`.

Reservoir pesticides, salts, constituents

`res_pest.f90`, `res_salt.f90`, and `res_cs.f90` run when those constituent modules are active. See Pesticides and Salts and constituents.

Wetlands on HRUs

Wetlands are configured per HRU via `wet_initial.f90` and `wetland.wet`. Each wetland has:

- Principal volume `wet_ob%pvol` (m^3) computed from `wet_hyd%pdep` (principal depth, mm) times the HRU area.
- Emergency volume `wet_ob%evol` (m^3) from `wet_hyd%edep`.
- Principal and emergency surface areas `psa` and `esa`.
- Optional weir geometry `weir_hgt` and `weir_wid` read from `weir.res`.

The wetland state `wet(iihru)` is the same `water_body` type used for reservoirs. In `hru_control.f90` the HRU surface runoff is routed into the wetland (`wet(j) = wet(j) + ht1`), the wetland runs its water balance and release (using the same `res_hydro` machinery on a wetland), and the outflow is added to the HRU outgoing hydrograph. Storage to depth uses a quadratic relation from `wet_hyd%bcoef` and `wet_hyd%ccoef`:

```
depth^2 * ccoef + depth * bcoef = (1 - flo / pvol)
```

with the positive root taken.

Wetlands run their own sediment, nutrient, pesticide, salt, and constituent routines using the same module code paths as reservoirs.

Switches and parameters

Switch / parameter	Default	File	Effect
`res_hyd%iyres`	per reservoir	`hydrology-res.res`	Year reservoir comes online
`res_hyd%mores`	per reservoir	`hydrology-res.res`	Month reservoir comes online
`pvol_m3`	per reservoir	`hydrology-res.res`	Principal volume (m^3)
`evol_m3`	per reservoir	`hydrology-res.res`	Emergency volume (m^3)
`wbody_prm%sed%nsed`	per reservoir	`sediment-res.sed`	Equilibrium sediment concentration (ppm)
`wbody_prm%sed_stlr_co`	per reservoir	`sediment-res.sed`	Sediment settling coefficient
`wbody_prm%weir_hgt`	per reservoir	`weir.res`	Weir height (m)
`wbody_prm%weir_wid`	per reservoir	`weir.res`	Weir width (m)

Switch / parameter	Default	File	Effect
`res_ob%release_dtl`	per reservoir	`dtl` (`res_rel.dtl`)	Decision table name for release rules
`wet_hyd%pdep`	per HRU wetland	`wetland.wet`	Principal depth (mm)
`wet_hyd%edep`	per HRU wetland	`wetland.wet`	Emergency depth (mm)
`wet_hyd%psa`	per HRU wetland	`wetland.wet`	Principal surface area fraction of HRU
`wet_hyd%esa`	per HRU wetland	`wetland.wet`	Emergency surface area fraction of HRU
`wet_hyd%bcoef`	per HRU wetland	`wetland.wet`	Storage-depth quadratic coefficient
`wet_hyd%ccoef`	per HRU wetland	`wetland.wet`	Storage-depth quadratic coefficient

TODO: verify default values in `hydrology-res.res`, `sediment-res.sed`, `nutrients-res.nut`, `weir.res`, and `wetland.wet` against their respective `_read.f90` readers.

Implementation

Source modules in `swatplus/src/`:

- `res_control.f90` is the master daily driver for a reservoir object. Adds inflow, computes the water balance, calls release, sediment, nutrient, pesticide, salt, and constituent routines.
- `res_hydro.f90` evaluates the release decision table and computes outflow. Houses weir, principal-volume, emergency-volume, Doell, Hanazaki, and HYPE release branches.
- `res_weir_release.f90` integrates weir discharge over sub-daily steps.
- `res_rel_conds.f90` evaluates release conditions specific to reservoirs.
- `res_sediment.f90` computes in-reservoir sediment settling.
- `res_nutrient.f90` runs in-reservoir nutrient transformations.
- `res_pest.f90` handles reservoir pesticides.
- `res_salt.f90`, `res_cs.f90` handle salts and constituents.
- `res_initial.f90`, `res_allo.f90`, `res_objects.f90` set up reservoir state at simulation start.
- `res_read.f90`, `res_read_hyd.f90`, `res_read_sed.f90`, `res_read_nut.f90`, `res_read_weir.f90`, `res_read_init.f90`, `res_read_conds.f90`, `res_read_elements.f90`, `res_read_csdb.f90`, `res_read_saltdb.f90`, `res_read_salt_cs.f90` read the reservoir input files.
- `reservoir_module.f90` defines reservoir types including `wet_ob` for wetlands.
- `reservoir_data_module.f90` and `reservoir_conditions_module.f90` hold parameter and conditions arrays.
- `reservoir_output.f90`, `res_pesticide_output.f90`, `res_salt_output.f90`, `res_cs_output.f90` write reservoir outputs.
- `wet_initial.f90` initialises a wetland on each flagged HRU.
- `wet_all_initial.f90` initialises all wetlands at simulation start.
- `wet_fp_init.f90` sets up the wetland-floodplain link to channels.
- `wet_read.f90`, `wet_read_hyd.f90`, `wet_read_salt_cs.f90` read wetland input files.
- `wet_irrp.f90` handles irrigation withdrawal from a wetland.
- `wet_cs.f90`, `wet_cs_output.f90`, `wet_salt.f90`, `wet_salt_output.f90` handle wetland constituents and salts.

Related pages

- Decision tables for how release rules are evaluated.
- `dtl` for the decision table input file format.
- `res` for the reservoir input files.
- `wet` for the wetland input files.
- Channels for downstream routing.
- Aquifer, reservoir, and plant output.